Arnava अर्णव
Vol. VIII No.1 Half Yearly Year 2019 ISSN 2320-0103

K. S. Chandra*

# A Study of Some Celestial Events Missed by Ancient Indians

**Abstract**

*Every 50 years or so, a star in our Galaxy blows itself apart in a supernova explosion, the most violent events in the Universe. The force of these explosions produces spectacular light shows. Explosions in past millennia have been bright enough to catch the attention of early astronomers hundreds of years before the telescope had, been invented. These type of Celestial Events itself suggests that the phenomenon occurring in Universe at a particular period, may or may not have been noticed by humans living in that era. However, whenever it was, noticed Ancient humans did record it in some or the other form, but some stellar events were, missed by Ancient Indians, as there is no record of it ever happened. In this paper, I am trying to put forward views and observations from earlier studies done by eminent Indian and Foreign Astronomers, Scholars and Scientists along with probable pictorial evidence of the mentioned events.*

**Keywords :** Nebula, Neolithic, Neutron Star, Pulsar, Supernova Remnant, White Dwarf

**सारांश**

लगभग प्रत्येक 50 वर्ष में हमारी आकाश गंगा से एक तारा सुपरनोवा विस्फोट में टूटकर अलग हो जाता है। इस विस्फोट की शक्ति के कारण अत्यन्त सुन्दर व शानदार

*Lecturer, Dept. of AIHC & Archaeology, RTM Nagpur University, Nagpur

रोशनियाँ दिखाई देती है। दूरबीनों के आविष्कार से भी पहले ये विस्फोट अपनी चमक के कारण खगोलशास्त्रियों को आकर्षित करते रहे हैं। इस प्रकार की आकाशीय घटनाओं से इस बात की संभावना बनती है कि यह बहुत संभव है कि उस काल के मनुष्य इन घटनाओं का साक्षात्कार नहीं कर पाते होंगे। जब भी ऐसी घटनाएं होती थीं प्राचीन काल का मानव उन्हें किसी न किसी प्रकार से संजो लेता था; हालांकि कुछ घटनाओं का कोई प्रमाण नहीं मिलता मानो वे कभी घटी ही नहीं थीं। प्रस्तुत शोधपत्र में विख्यात भारतीय व विदेशी खगोलशास्त्रियों, विद्वानों और वैज्ञानिकों के विचार और अवलोकन के साथ ही इन घटनाओं के चित्रमय प्रमाणों को प्रस्तुत करने का मेरा प्रयास है।

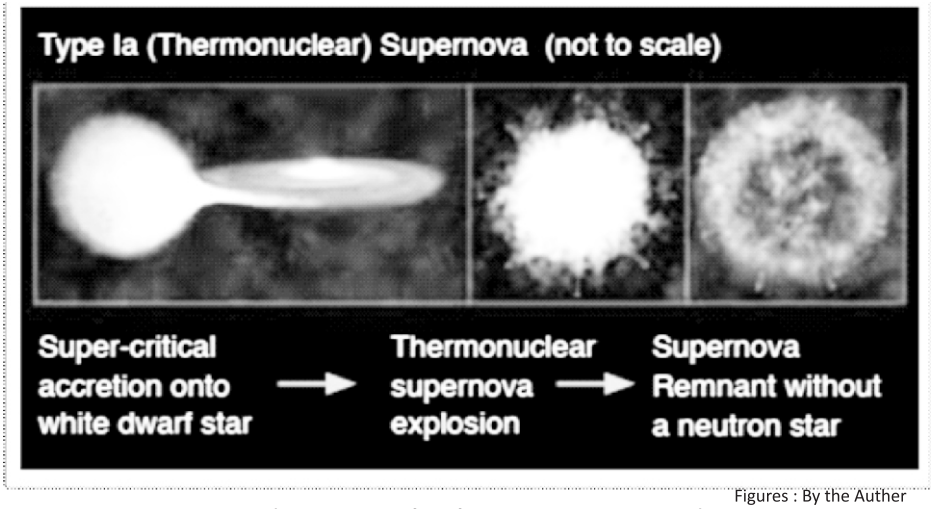
**प्रमुख शब्द** : नेब्यूला, नवपाषाणिक, न्यूट्रॉन तारा, पल्सर, सुपरनोवा, अवशेष, श्वेत बौना

## Introduction

Ancient Indian History consists of various components like Vedas, Upanishads, Epics, Samhitas, Ballads, and Rituals along with Buddhism and Jainism, where in most of the celestial bodies have been given human form and worshipped across the length and breadth of the country, Sun (Surya), Moon (Chandra), Constellations (Rashi), Nine Planets (Nav Graha), Stars (Nakshatra). Epigraphic, Numismatics and Iconographic evidences coming from all the locations in India prove this. Even in modern times, Indians worship these deities on specific days of the month in specially erected individual temples a tradition which is been followed from generations to generations. Such a traditional and Celestial body worshipping Indian society did miss the stellar events, which occurred between 4500 BCE - 1200 CE and aptly being recorded by Arab, Chinese, European and North American Astronomers of that era (Joglekar *et. al.* 2006:1-7).

## About Supernova

Supernova is an extremely violent explosion that marks end of stellar life. Supernovae are among the brightest events in the Universe. Some supernovae are so bright that they can outshine the entire galaxy.




Figures : By the Auther

**Fig.1 : Type-Ia Supernova-When accretion of gas from a companion star in a binary system increases the mass of a white dwarf star beyond 1.4 solar masses, temperatures in the core of the white dwarf become high enough for carbon fusion to occur. Fusion begins throughout the star almost simultaneously and a powerful explosion occurs, leaving no remnant.**

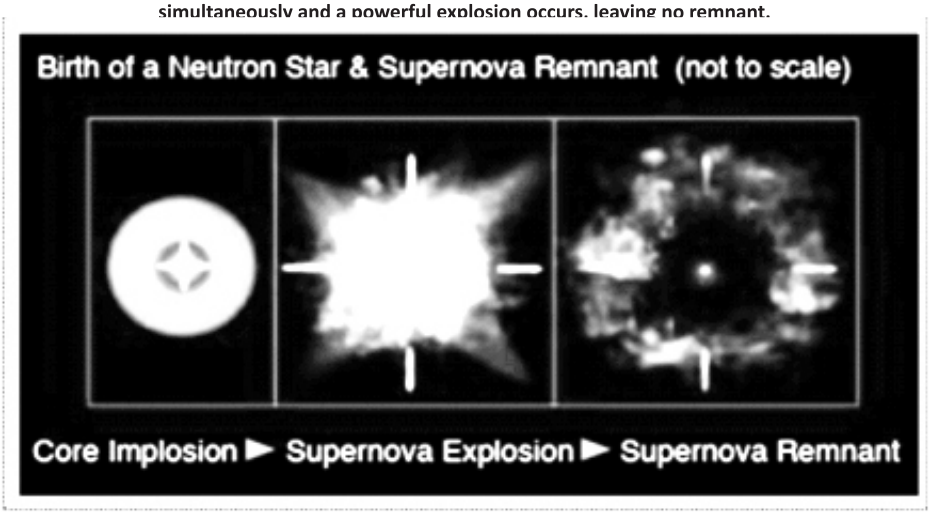



**Fig.2 : Type-II Supernova-When the nuclear power source at the core of a massive star is exhausted, the core collapses. In less than a second, a neutron star (or a black hole if the star is extremely massive) is, formed. An enormous amount of energy is, released, reversing the implosion. All but the central neutron star is, blown away at tremendous speeds, accompanied by a shock wave that races through the expanding stellar debris.**

There are two basic types of supernova:

**Type Ia** (Fig.1) happens when a white dwarf star draws large amounts of matter from a nearby star, until it can no longer support itself, and collapses.

**Type IIa** (Fig.2), is the result of the collapse of a massive star.

These two types of supernovae can be, uniquely identified by their intensity curves. At the time of supernova, brightness of star can instantaneously increase to more than a million times its original brightness. It can remain in such a state for a few hours and after that, it fades away slowly over next few months. Hence, if a supernova occurs in our vicinity, it can appear so bright that it will outshine most of the objects in the night sky. In rare cases, it can even outshine the moon and be visible in daylight. However, supernovae are among the rarest events to observe. Last two millennia may have seen few supernovae. Supernovae sightings have been, recorded throughout history although the distinction of true supernovae from novae is sometimes ambiguous.

**The Events**

Until now, there has been no epigraphic or iconographic recorded evidence of stellar event happening in the Ancient Past except from the Neolithic Settlement of Burzahom in Kashmir, North India.

**Years between 4500-4200 BCE**

The first and only presumed to be the recorded event in India comes from Neolithic Settlement of Burzahom dating back to about 4325(+/-115) BCE (Sharma, 2000), a stone carving was unearthed by (Khazanchi, 1969:13) (Fig. 3). The carving is on an irregular stone slab with size of about 48 cm by 27 cm. The plate is, chipped at one corner, leaving a possibility that a part of the carving may be missing. The slab was, recovered from phase IB of the site re-utilized for a larger structure with inscribed surface facing interior of the structure (Pande, 1971:134-138; Agrawal and Kusumgar, 1965: 42-43). The figure shows two bright objects in the sky with rays of light coming out of them and a hunter spearing an animal below the first object. There is another animal to the left of the

hunter drawn above the hunter's spear. It is, suggested that it represents a hunting scene and the two objects are pair of bright stars at the local zenith at the beginning of the hunting season. Few such candidate pairs



**Fig.3 : Photograph of Stone Carving from Burzahom (courtesy-IGNCA, along-with-a-sketch-of-the-same)**

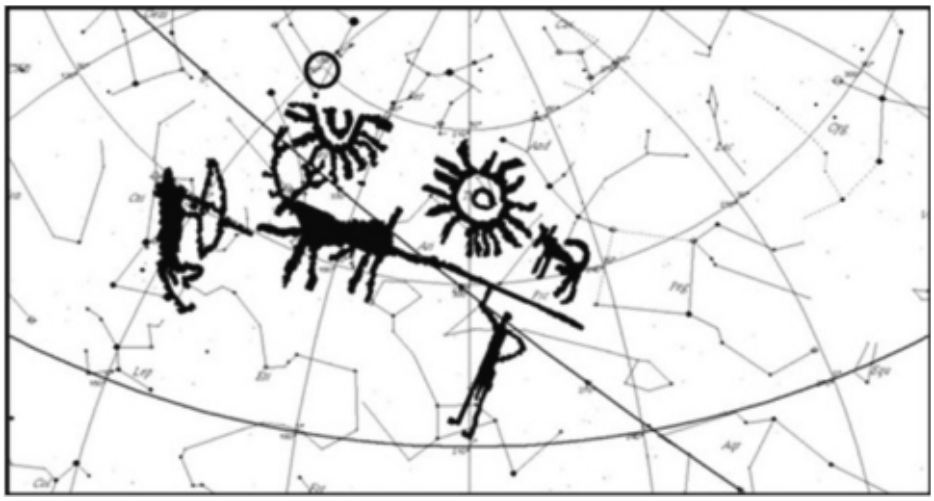


**Fig.4 : Sky Map of the Region of HB9 in the Sky Chart for 5700 BCE**

of stars have been, suggested (Joglekar et. al 2006:1-7).

Scholars investigated the possibility that the rock drawing is the record of the supernova HB9 ((Joglekar et.al. 2006:1-7) and suggested that the partially drawn object is HB9 since it would be irregular and that the second bright object is Moon since the apparent magnitude of HB9 is closer to that of the Moon. It was, observed that this was not a terrestrial hunting scene but is actually a sky-map giving location of prominent

constellations and the Moon on the day the supernova was first seen (Fig.4). The striking similarity of the patterns drawn on the sky map with the original drawing is clearly visible. On the sky chart the image of one of the hunters coincides with the Orion; the central stag is same as the Taurus. The hunter on the right may have, been formed from stars of Cetus and other animal on the right may be Andromeda and Pegasus. The long, curved line in the carving, traditionally interpreted as spear, may well be an arc of bright stars. These sky patterns account for all the bright stars in the region and look consistent with then prevalent culture.

As the constellations have iconographic importance in primitive, ancient, medieval and continuing into modern cultures. The probable



**Fig.5 : The Probable Supernova Remnant G160.9 +02.6/HB9/Sh2 221 Depicted at Burzahom Neolithic site**

remnant of the supernova HB9 has been recorded recently (Fig.5).

Supernova Remnant G160.9 02.6, HB9, or Sh2 221 estimated to be less than 4 kpc distant and expanding at an estimated 18km/sec. To the

right or south of G160.9 +02.6 is LD1475, 76, 77, 80 VDB32 a clear absence of stars and a significantly large H-Alpha emission nebula covering the same area, near its center is the bright star Almaaz-Epsilon

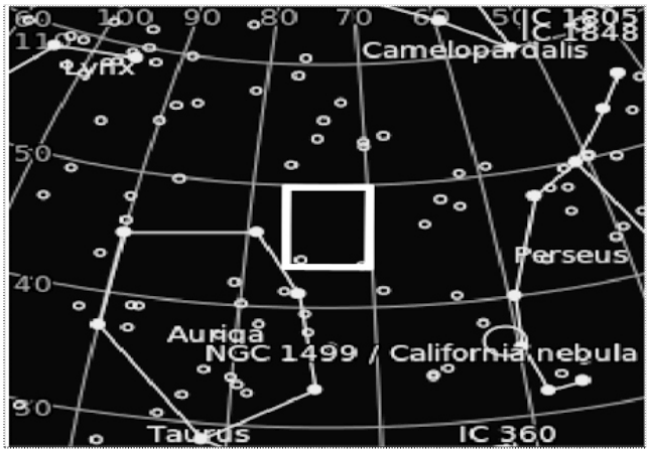



**Fig.6 : Sky Map Depicting HB9**

Aurigae (www.astrobin.com) (Fig.6).

**Year 185 CE**

A Supernova exploded in the constellation of Centaurus, roughly between α Centauri and β Centauri in the year 185 CE. The peak

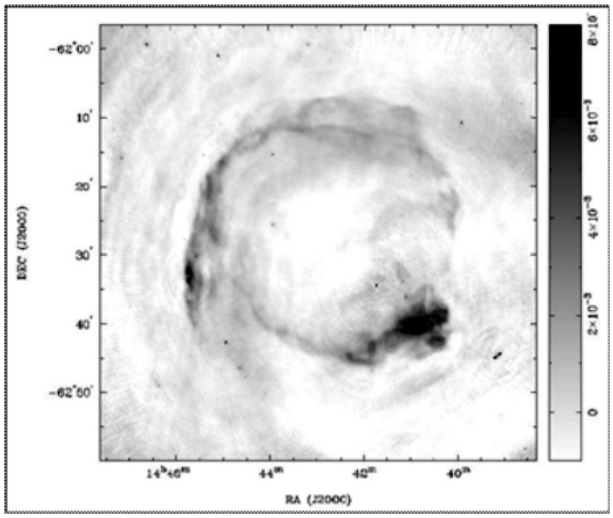



**Fig.7 : Radio Emission- A Grayscale Image G315.4-2.3 from Australia Telescope Compact Array (ATCA) observations at 1.34 GHz (After Michael Bietenholz)**

brightness of supernova was, estimated to be brighter than Sirius (Vyadh). As per Chinese astronomical records, this supernova was visible for more than 8 months. Centaurus lies in the southern skies, westward of the constellation Scorpius. In Indian mythology, stars α Centauri and β Centauri are, known as Mitra and Mitrak, respectively. We can see a shell of radio emission called G315.4-2.3, which probably may be the remnant

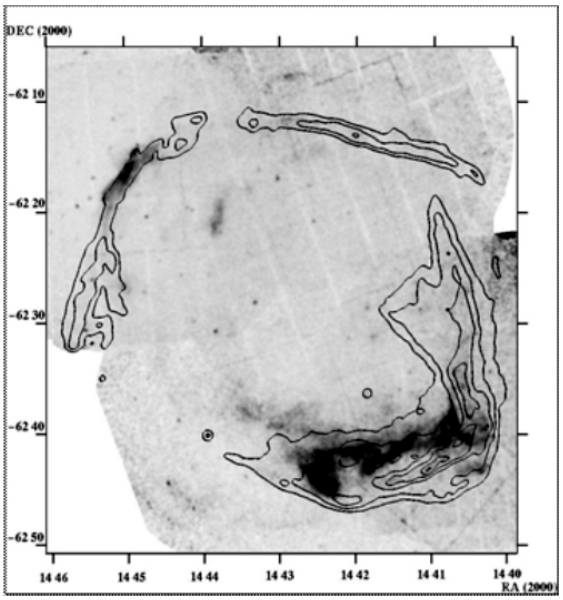



**Fig.8 : An XMM-Newton X-ray Image of G315.4-2.3**

of this supernova & Fig.7 & Fig.8.

As reported in a single Chinese record, the supernova of 185 CE was visible for at least 8 months and reached brightness comparable to Mars. Optical, radio, and X-ray emission observed at a location consistent with the Chinese record make G315.4-2.3 the prime candidate for the remnant of SN 185 AD. Combined images from the Chandra and XMM-Newton X-ray observatories show low, medium and high-energy X-rays in red, green, and blue respectively. By studying the amount of X-rays at different energies, and measuring the Remnant's size, scientists now surmise that, the explosion of a massive star created it roughly 2,000

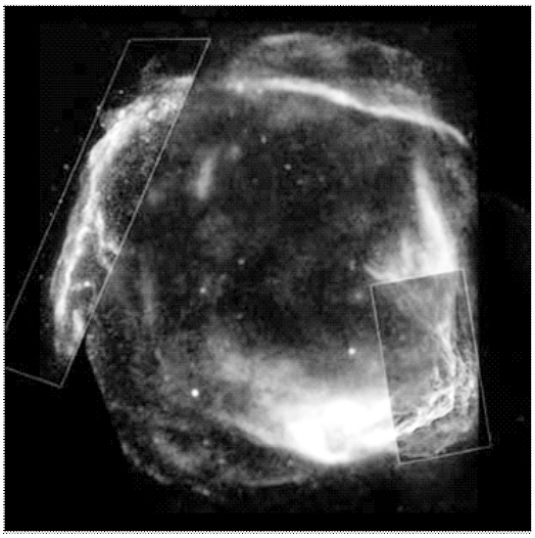


**Fig.9 : Combined Images of G315.4-2.3 from the Chandra and XMM-Newton X-ray Observatories**

years ago(Chandra: NASA/CXC/Univ. of. Utrecht/J.Vink et. al.) (Fig.9).

**Year 393 CE**

This supernova appeared in the tail of Scorpion according to two different Chinese texts in the year 393 CE. However, the exact date is difficult to estimate. The supernova was between Jupiter and Venus in brightness and later faded away in the next 7 months. Three radio emission sources had been, identified in this region, but none of them is conclusive enough to be, identified as supernova remnant (Joglekar et. al. 2006: 1-7). NASA Chandra observatory has, probably identified it as G347.3-0.5. There are several supernova remnants within this region, so it is difficult to identify the remnant of SN 393 CE with certainty. X-rays from G347.3-0.5 are, dominated by radiation from extremely high-energy electrons in a magnetized shell rather than radiation from a hot gas. The remnant, seen by Chandra (inset box) and XMM-Newton, is also a source of very high-energy gamma rays. The bright, point-like source on the lower right in the image (which shows only the upper portion of the entire remnant) (Fig.10) is similar to other known neutron stars and

indicates that G347.3-0.5 is the remnant of a core-collapse supernova (Chandra: NASA/CXC/SAO/P.Slane et al). However, scientists like (Robert A. Fesen et. al. 2012: 1-6) are skeptical about SN 393(G347.3-0.5) being

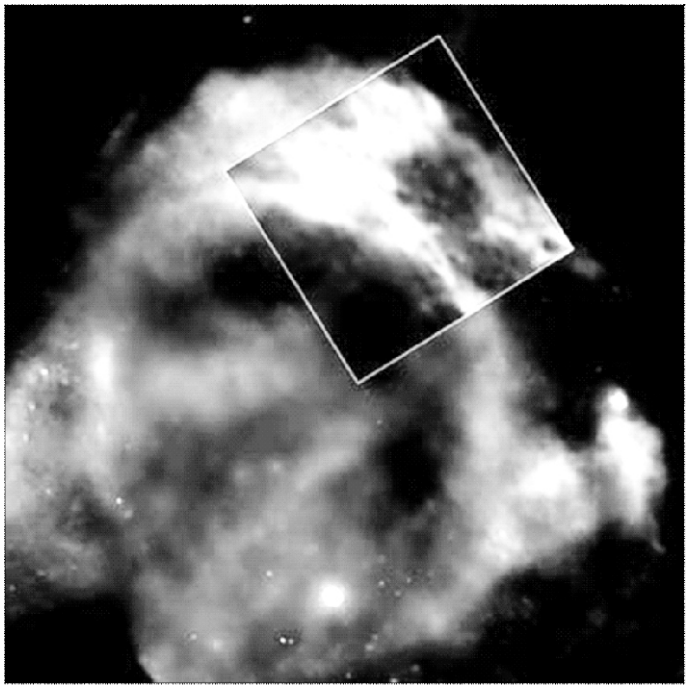


**Fig.10 : Probable Remnant of SN 393**

the same as identified in ancient Chinese and Roman accounts.

**Year 1006 CE**

Supernova of the year 1006 is undoubtedly the brightest of all recorded supernovae. It was, noted across most of the civilizations in the world -, China, Egypt, Europe, Iraq, Japan and Korea. Supernova blazed 100 times brighter than Venus did. The supernova was, seen near Lupi; probably seen first on April 30, 1006. Lupus lies westward of the constellation Scorpius. In 1960s, radio astronomers identified remnant of SN1006, technically called as G327.6+14.6 making it the brightest supernova recorded as seen from Earth. A number of lines of evidence its location well above the Galactic disk, the detection of significant

amounts of cool iron gas in its interior. Absence of any evidence for a neutron star, and the implication from the historical record that it remained visible for several years, imply that SN 1006 was the result of the explosion of a white dwarf star (Credit: NASA/CXC/Rutgers/J.Hughes et. al.) (Fig.11).

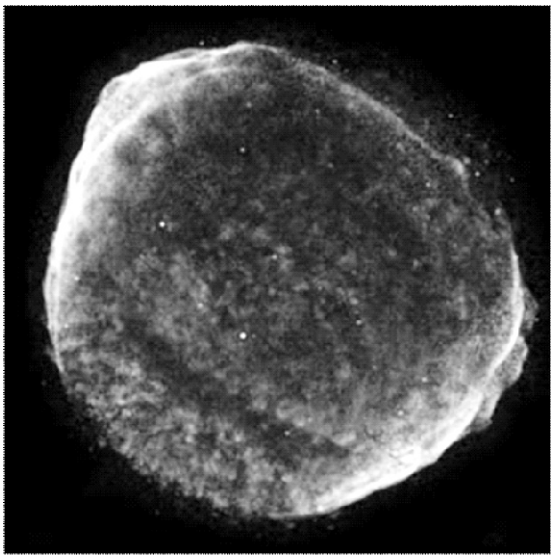


**Fig.11 : Chandra Image of SN1006 Shows X-rays Produced by High-Energy Particles (blue) and Multimillion-Degree Gas (Red/Green)**

**Year 1054 CE**

The most popular historical supernova was supposed to have exploded on July 4, 1054, in the constellation of Taurus. References to this supernova can be, found in well-documented Chinese records to inscriptions of Native Americans (Fig.12). It is probably the nearest supernova. It was visible in early morning skies when it exploded. Crescent of the moon was very close to the supernova. The guest star was visible in the daylight for about 23 days. It was visible for the next 20 months in the night skies before it faded out. The nebulous remnant was, discovered by John Bevis in 1731 and was, discovered independently by Charles Messier. This remnant is known as M1, being the first object in

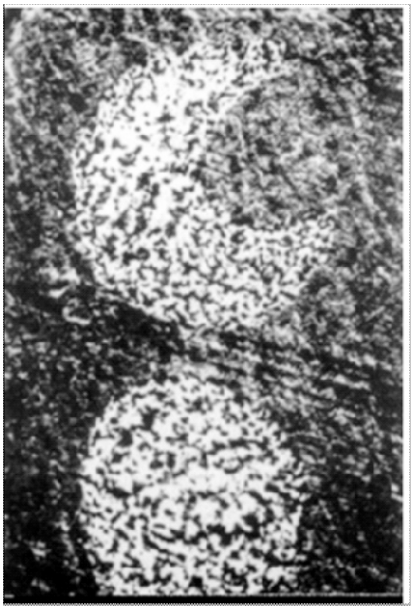


Fig.12 A

Fig.12 A : Pictograph in the Navajo Canyon Area in North America Showing Crescent Moon along with a Bright Star-like Object



Fig.12 B

Fig.12 B : Petroglyph in the White Mesa Region, which also Indicates the Same Event. [Both pictures are ascribed to Pueblo Indians of the eleventh century (Narlikar and Bhate 2001: 702)]

Messier's list. The remnant is, also known as the famous Crab Nebula.

In the History of Sung Dynasty by Ho Peng Yoke, published in 1962 the following event is, described: “On a Chi-Chhou day in the fifth month of the year of Chi-Ho reign period, 'a guest star' appeared at the south-east of Tien-Kaun measuring several inches. After more than a year it faded away”. It was in that context that the above event was, noticed and duly recorded as, stated above, and 4 July 1054 is the date, according to the modern calendar, corresponding to the Chinese record. The Chinese had the custom of describing such transient objects as guests in the sky. The sighting of this object was, also recorded in Japan, where too the astrologers kept meticulous records of the heavens. The recorded direction of the object, points to Tauri in the constellation of the 'bull'. What do we see there today is the Crab Nebula/M1. The same remarkable sight was, seen and recorded in the Middle East by a Christian

physician from Baghdad, named Ibn Butan. Although not a professional astronomer or astrologer, Ibn Butan, like his contemporary physicians, was interested in the possibility that diseases on the earth could be, related to cosmic events. Ibn Butan's biography was, recorded in a biographical encyclopedia prepared by Ibn Abi Usaybia around 1242 AD, in which his report is, reproduced. Some extracts, from that report, are illuminating.

*"... One of the well-known epidemics of our own time is that which occurred when the spectacular star appeared in Gemini in the year 446H. In the autumn of that year, fourteen thousand people were, buried in the Church of Luke, after all the cemeteries in Constantinople had been, filled... As this spectacular star appeared in the sign of Gemini... it caused the epidemic to break out in Fustat when the Nile was low, at the time of its appearance in the year 445H (sic)..."*

Here, the year is measured on the Islamic Hizri calendar, according to which the year 446H corresponds to the period 12 April 1054 to 1 April 1055 CE, that is encompassing the dates when the Chinese saw the guest star. Ibn Butan seems to imply that this event occurred in the summer and caused the epidemic in the following autumn, when the Nile was low. This places the event in the summer of 1054, which agrees with the more precise Chinese date of 4 July 1054 CE. One additional point may be noted. The Crab Nebula is, seen today in the constellation Taurus, whereas Ibn Butan refers to Gemini. If he were using the astrological convention, he would give the location with respect to the sun's location in the zodiac. If one takes into account the steady precession of the earth's rotation axis, the Crab Nebula would have appeared in Gemini about a thousand years ago (Narlikar and Bhate 2001:701-2).

We have a speculation about the supernova. In Indian mythology,

♌Tauri is given the name Agni (The god of Fire). ♌Tauri is a very ordinary star and it is difficult to imagine why such an aggressive name is, given to an ordinary star. In fact, ♋Tauri (Rohinee) could have been a very close candidate for a name Agni due to its red color. We speculate that the name originally given, to the 'new star' that was shining with brightness 5 times greater than that of Venus. After supernova faded away, name was stuck with ♌Tauri being brighter than ♋Tauri. However, this speculation needs to be, crosschecked by searching for a possible reference to the star ♌Tauri being, designated as Agni before 1054 CE; on the onset, there is also strong evidence to show that the supernova of 1054 CE was the




**Fig.13 : The X-ray Image, taken by the Chandra X-ray Observatory, Shows Two Rings of Material around the Crab Pulsar (Credits: Chandra X-Ray Telescope image: NASA and ESA)**

explosion that produced the Crab Nebula. The bright nebula, which has been, known optically since the early 18th century, is due to the activity of a neutron star (bright white dot in the center of the image) (Fig.13) that is rotating 30 times a second and spewing out a blizzard of extremely high-energy particles. Chandra's image of the Crab Nebula in Taurus Constellation 6,500 Light Years away reveals rings and jets of high-energy particles that appear to have been, flung outward over great distances from the neutron star (Fig.14). The diameter of the inner ring is about 1,000 times the diameter of our solar system (Credit:

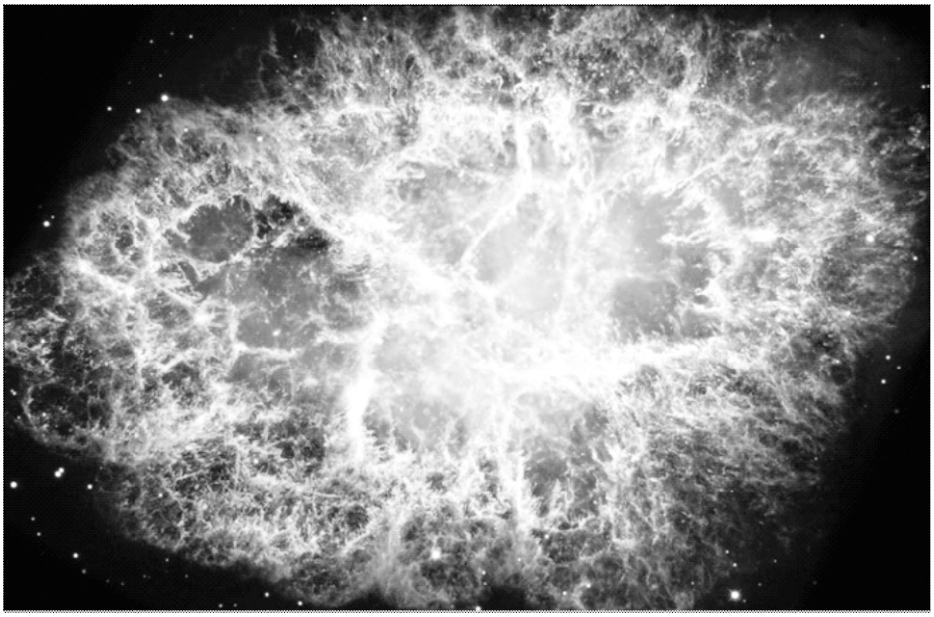


**Fig.14 : Crab Nebula Picture taken in Visible Light by the Hubble Telescope, shows the Pulsar's Location at the Heart of the Nebula. The Pulsar cannot be seen in Visible Light (Credits: Hubble image: NASA and ESA)**

NASA/CXC/ASU/J.Hester *et. al.*).

The Nebula with its direction matching that of the ancient Chinese records, and its physical environment consistent with the remnant of what that event was, astronomers are sure that the Guest star did not actually go away, but still around in the shape of its remnant the Crab Nebula. The approximate distance of this nebula from us is 6,500 light years, while the extent of the whole structure of is as large as 5 to 10 light years. (One light year is the distance Travelled by light in one year, and is approximately ten million millions of kilometers.). Therefore, this is (Fig.14) what remains today of the event that was witnessed by the Chinese, the Red Indians and the Middle Easterners, nine and a half centuries ago. The event is a catastrophic one, marking the break-up of a star (Narlikar and Bhate 2001: 702).

**Year 1181 CE**

A Supernova exploded in northern skies in 1181 CE. Its peak magnitude is, estimated to be–1 (slightly dimmer than Sirius). It was, mainly observed by Japanese and Chinese civilization. Supernova was visible in constellation of Cassiopeia (Sharmishtha) for 185 days. Astronomers with NASA's Chandra X-ray Observatory have found a pulsating star only 10,000 light years from Earth. The object discovered by the astronomers is a type of whirling neutron star known as a pulsar, the remnant of a once bright and burning star. First identified by Asian astronomers in the year 1181, its reported position is close to the supernova remnant pulsar 3C58 (Plain: Kennedy Space Center). The discovery of a rapidly rotating neutron star (inside the bright spot in the center of the image) (Fig. 15) in the remnant, and the overall similarity in appearance to the Crab Nebula argue in favor of identification consistent with an age of somewhat less than 1000 years. However, the rate at which the filamentary shell of gas is expanding is much too slow to have reached its present size in 826 years, so the association of 3C58 with SN 1181 is, listed as possible rather than definite (Credit: NASA/CXC/SAO/P.Slane *et. al.*).

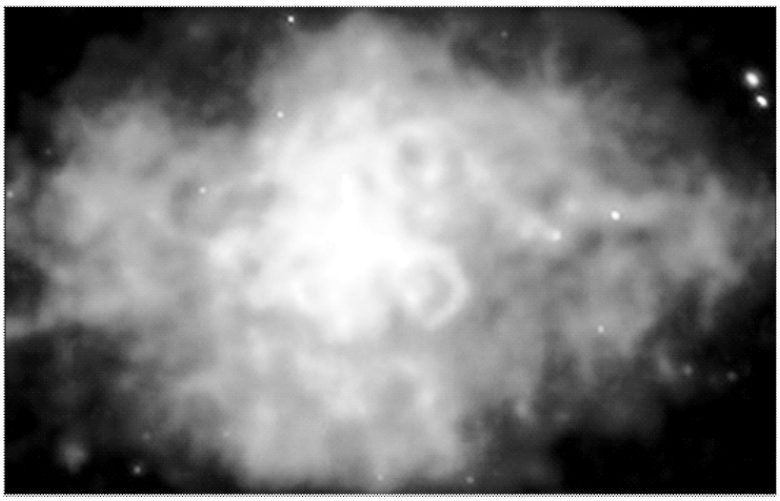


**Fig.15 : The Image of Pulsar 3C58/SN1181 taken by the Chandra X-ray Observatory shows the Glow of Radiation Contained within the Star's Magnetic Field (Credit: NASA/CXC/SAO)**

## Conclusion

We thus have different sources of information about the sighting of a unique cosmic event, from China and Japan in East Asia, from the American continent in the western hemisphere, as well as from the Middle East in West Asia. Why no records from India?., In India, astronomy was flourishing in the golden era which had started with Aryabhata in the 5th Century and which was to continue until Bhaskara II in the 12th Century. Such an event would have been, witnessed by many at least, in some part of the subcontinent, and when something unusual was seen, the astronomers and astrologers would have been, consulted. Many people in India must have observed such a brilliant event and it is hard to imagine that no records of these sightings were, maintained. They must have therefore recorded some events. The explanation may be that there was not much written tradition in India at the time, the emphasis of scholarship being on reading the ancient texts, rather than creating new ones (Narlikar and Bhate, 2001:701-3). However, Indian method of recording is far more complex with an insistence on brevity and mysticism. Records of Solar and Lunar eclipse are, found in South Indian inscriptions but no mention of other cosmic events (Shylaja and Kydala, 2011: 335-343). Epigraphic sources mostly give us the information about Day, Date, Thiti, Varsa (mainly regnal years of the issuer), Lineage Solar/Lunar (Suryavamsi/Chandravamsi), Rashi (Zodiac Sign), Gotra and Nakshatra in the Copper Plate/Stone/Rock/Pillar /Temple/Stupa/Chaitya inscriptions. However, no mention of the cosmic events in any of the scriptures found until now. The, society being superstitious has not recorded these stellar events is still an intriguing mystery or if might have been recorded and kept as dark secrets which might be waiting to uncover.

**View from the Above Discussion**

- This kind of celestial event was never recorded

**OR**

- The records destroyed/pilferage/looted by foreign invaders

**OR**

- The records are hidden and waiting to see the light of the day by the way of chance discovery or Archaeological Excavation.

  Until then the puzzle remains unsolved

**Acknowledgement**

My Sincere thanks to Dr. Preety Trivedi Professor and HOD and Dr. Prabash Sahu Associate Professor, Dept. of Ancient Indian History Culture and Archaeology; RTM Nagpur University, Nagpur for their support and motivation in preparing this article.

**References**

- Abbas Riza Vahia Mayank, Shobha.V, Mohanty Ravi Kumar and Sule Aniket: 2010, Project Report "Search for Archaeo-astronomical Data from the Megalithic Burials of Junapani, Nagpur, Maharashtra", Indian Rock Art Research Centre, Nasik (A division of Indian Numismatic, Historical and Cultural Research Foundation) and Department of Astrophysics, Tata Institute of Fundamental Research, Mumbai
- Agrawal, D.P and S. Kusumgar: 1965, Current Science, Vol.34
- Bietenholz, Michel: 1740, The Radio Spectral Index of the Supernova Remnant G315.4-2.3, MSc. Project, Hartebeesthoek Radio Astronomy Observatory PO Box 443, Krugersdorp
- Blasts from the Past: Historic Supernovas, NASA's Marshall Space Flight Center, Huntsville, Alaska Manages the Chandra program

for the agency's Science Mission Directorate, The Smithsonian Astrophysical Observatory controls science and flight operations from the Chandra X-ray Center in Cambridge, Mass :1-2

- Joglekar, Hrishikesh & Aniket Sule, M. N. Vahia: 2006, In search of Indian records of Supernovae, Indian Journal of History of Science, 1-7
- Khazanchi, T. N.: 1969, Explorations and Excavations in Jammu and Kashmir, Archaeological Survey Of India, 1964-65, Ed. A Ghosh
- Narlikar, Jayant V. and Bhate Saroja: 2001, Search for Ancient Indian Records of the Sighting of Supernovae (Historical Notes), Current Science, Vol. 81, No. 6(25)
- Pande, B. M.: 1971, Asian Perspectives 14
- Sharma, A. K.: 2000, Early Man in Jammu Kashmir and Ladakh, Agam Kala Prakashan, Delhi
- Shylaja, B. S. and Geetha Kydala: 2011, Inscriptions as Records of Celestial Events, Indian Journal of History of Science, 46(2)
- : 2016, Records of Vyatipata in Stone Inscriptions, Indian Journal of History of Science, 2016, 51(2.1): 206-216 DOI: 10.16943/ijhs/2016/v51i2/48432. The Crab Nebula (M1), National Aeronautics and Space Administration, Goddard Space Flight Centre, 8800 Greenbelt Road Greenbelt, Maryland 20771, USA

**Further Readings**

- Burgess, James: 1893, Notes on Hindu Astronomy and the History of our Knowledge of it, Journal of the Royal Asiatic Society of

Great Britain and Ireland, London, Art XVIII: 717-761

- Damashek M. & J. H.Taylor R. A.Hulse: 1978, ApJ, 225, L31-33 Fesen Robert A, Kremer Richard, Patnaude Daniel and Milisavljevic Dan, 2012, the Sn 393–Snr RxJ1713.7-3946 (G347.3-0.5) Connection, The Astronomical Journal, 143(27): 1-6, doi:10.1088/0004-6256/143/2/27
- Henshilwood, C. S. & F. dOErrico, R .Yates, Z .Jacobs, C .Tribolo, G. A. T.Duller, N.Mercier, J .Sealy, H. Valladas, I .Watts, A. G.Wintle: 2002, Science, 295: 1278

**Webliography**

- https://www.astrobin.com/229090/B/?nc=all
- http://chandra.harvard.edu/chronicle/0406/historical_snr/
- http://www.crystalinks.com/indiastronomy.html
- www.mrao.cam.ac.uk/surveys/snrs/snrs.data.html
- www.nasa.gov
- www.vepa.us
- Astronomy in Ancient India, Indic Studies Foundation, (a California Non–Profit organization) www.indicethos.org, www.indicstudies.us

★★★